IQP Generative Models for Strong Gravitational Lensing Images

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Abstract

Instantaneous-quantum-polynomial (IQP) circuits are believed to be hard to sample classically [4, 5], while the expectation values a maximum-mean-discrepancy (MMD) loss needs stay computable, so an IQP Born machine can be trained classically and deployed on quantum hardware [15]. A recent correlation–complexity map, two indices computed from a dataset's Walsh order spectrum and its multi-information, predicts which datasets suit such models, and on it DeepLense strong-lensing images are the most IQP-compatible dataset we measured ($I_{\mathrm{QCLI}} = 0.47$, $I_{\mathrm{CCI}} = 0.87$), above the turbulence data of the paper that introduced it. Given *learned* latent directions, $\theta = \theta^0 + \Phi^\top z$, a 10-qubit IQP Born machine trained on lens crops beats an RBM and a DCGAN on the paper's PDF–JS and on flux-aligned Conv2D-MMD^2 at $N = 11$ and $N = 100$ training images with 2–39$\times$ fewer parameters, a different arm winning each metric, and reconstructs unseen images at correlation 0.97. Three caveats bound it. The reference paper's own fixed latent protocol yields only noise on images, a structure-matched permutation null still scores $I_{\mathrm{QCLI}} = 0.42$, and the DCGAN still leads on raw Conv2D-MMD^2 at $N = 100$. Everything is exact classical simulation at a single seed and $n = 10$ qubits, so we claim no quantum advantage.

1 Introduction

IQP circuits are commuting-gate circuits whose output distribution is the squared Walsh–Hadamard transform of a phase function. Exact classical sampling from them would collapse the polynomial hierarchy [4], and the hardness is conjectured to survive approximate sampling in the average case [5, 7], though sufficient noise restores classical simulability [14]. The same commuting structure keeps an MMD loss's expectation values computable, so such a model can be *trained classically and deployed on quantum hardware* [15, 16].

Whether real data sits where that is useful is an empirical question, and a recent proposal [11] approaches it with two ingredients. (i) A *correlation–complexity map* placing a dataset on two coordinates, the quantum correlation-likeness indicator I_{QCLI}, the Jensen–Shannon divergence between the Walsh order spectrum of the data and the i.i.d. binomial baseline $b_k = \binom{n}{k}/2^n$, and the classical correlation-complexity indicator I_{CCI}, how much of its total correlation the best pairwise tree misses [19, 17, 6]; the high-I_{QCLI}, high-I_{CCI} corner is argued to be where IQP circuits do well. (ii) A latent-adaptation protocol (Algorithm 1 of the reference paper, which trains a shared phase core on one anchor sample, then adapts a fixed block of 50 Walsh-basis phases taken in Hamming order sample by sample, and interpolates that block to reach unseen samples). Both were demonstrated on one dataset, 2D turbulence snapshots [10] at $n = 18$ qubits.

We take both to simulated dark-matter strong-lensing images from DeepLense [1, 2]. Lens crops land at the most IQP-compatible position of any dataset we measured, a 10-qubit IQP Born machine

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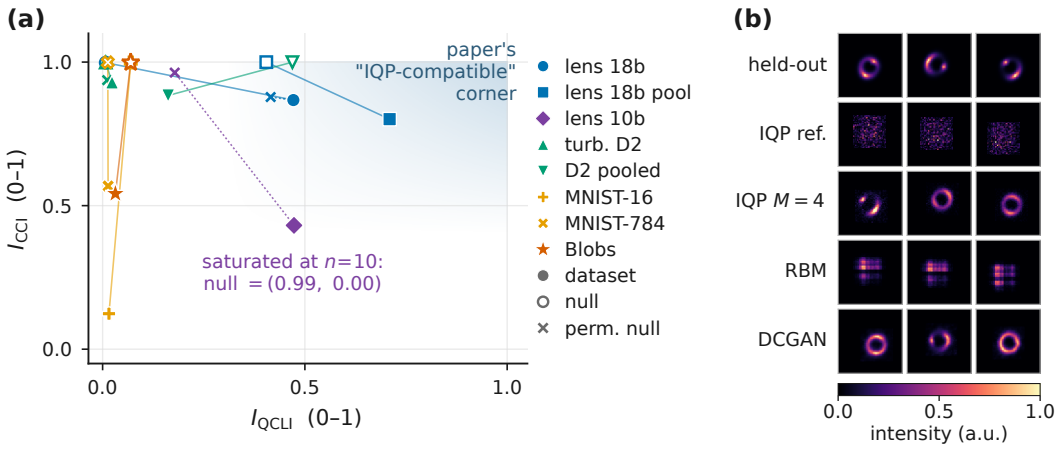


Figure 1: **(a)** Correlation–complexity map, with datasets filled, i.i.d.-uniform nulls at the same (n, K) open and labelled null, and permutation nulls as crosses labelled perm. null (ours, not in the reference paper). The legend also shortens lens 18b for the 18-bit per-image encoding, lens 18b pool for its pooled variant, lens 10b for the 10-bit crop pmf, turb. D2 and D2 pooled for the reference paper's turbulence data, MNIST-16 and MNIST-784 for MNIST binarised after mean-pooling to 4×4 and at its full 28×28, and Blobs for BinaryBlobs. **(b)** First three unconditional samples per model, $N{=}100$, no selection, shared 0–1 colour scale, with IQP ref. the reference protocol.

trained on them beats the paper's own RBM and DCGAN baselines on both of its metrics with far fewer parameters, one arm for each, and it does so only once the fixed Walsh latent block of the reference protocol is made *learnable*. Appendices B–D take the same tests back to the paper's own data, where its turbulence result does not reproduce under our pre-registered convention and two internal claims come out mixed.

2 Setup

Model. We compare the reference protocol against its fixed 50-phase block trained *jointly* across images and against our *learned-direction* model, a trainable Φ whose M rows are directions in phase space, so that each image is a point z_i in M dimensions,

$$\theta_i \;=\; \theta^0 + \Phi^\top z_i, \qquad \theta^0 \in \mathbb{R}^{1024}, \quad \Phi \in \mathbb{R}^{M\times 1024}, \quad z_i \in \mathbb{R}^M, \tag{1}$$

with $M \in \{2, 4, 8\}$, each row of Φ L^2-normalised to $\sqrt{1024}$, and an L^2 penalty on z of 10^{-3} raised to 0.3 for the $M{=}8$ arm. The objectives are the paper kernel (the reference paper's multi-bandwidth Gaussian MMD kernel), the flat MMD (uniform weight on every Walsh coefficient, which makes it estimable from hardware shots) and exact KL (a simulation-only diagnostic, App. E), and z is drawn from a fitted Gaussian or resampled with jitter.

Baselines and metrics. The baselines are a restricted Boltzmann machine (RBM; 10 visible and 1,000 hidden units, persistent contrastive divergence with one Gibbs step [9, 18]) and a deep convolutional generative adversarial network (DCGAN) [13], sized in Table 2. We keep the reference paper's two metrics, the probability-density Jensen–Shannon divergence between intensity histograms (PDF–JS) and Conv2D-MMD^2 in the feature space of a fixed random convolutional encoder [8], and add a held-out projection correlation. Every IQP arm and the RBM share the flux scalar S while the DCGAN learns its own scale, so raw Conv2D-MMD^2 is partly a brightness comparison and we also report a flux-aligned variant, flagged as post-hoc, and every arm generates 200 images scored against the same 200 held-out images at $N \in \{11, 100\}$ with seed 0 throughout.

Departures from the reference setup. Six choices differ from the reference paper, each following from the data. Its snapshots carry a time index and its task is to interpolate to unseen times, whereas the DeepLense `Model II` images we use [1, 3, 12] are static and fall into three substructure classes, so we generate unconditionally from the latent prior. It encodes a sample as 18-bit $(x, y, \text{intensity})$

Table 1: Correlation–complexity map on our data, each permutation null keeping the value histogram of the row above.

Dataset and encoding	K	I_{QCLI}	I_{CCI}
Lens, 18-bit (x, y, z) tuples	4,096	0.4715	0.8677
permutation null	4,096	0.4154	0.8782
Turbulence D2, 18-bit tuples	4,096	0.0232	0.9294
permutation null	4,096	0.0113	0.9371
Lens, 10-bit crop pmf	exact	0.4731	0.4313
permutation null	exact	0.1782	0.9633

tuples on 18 qubits, whereas we centroid-align each 64×64 image, crop the central 32×32 window holding 99.4% of the light and read it as one pmf over $1{,}024$ cells on $n = 10$ qubits (App. A), because a raw intensity axis carries barely more than one bit. Its sequential anchor-and-adapt protocol (Sec. 1) fixes the latent block, whereas we learn Φ and train the N images jointly, since fixed Walsh directions cross-fade pixels instead of moving a ring (Sec. 3.2). It estimates its MMD from 500 model samples per step, whereas we evaluate it in closed form, because a 10-qubit pmf is exactly computable. It measures generalisation by synthesising unseen times, whereas static images give no time axis to hold out, so we use a projection correlation on held-out images. We paste each generated crop back into 64×64 at a fixed flux $S = 124.92$ and a sampled offset, because scoring has to return to the original image scale.

3 Results

3.1 Lens images are the most IQP-compatible dataset on the map

Table 1 recomputes the map. Encoded as the reference paper encodes turbulence, lens images sit far above it in I_{QCLI} and inside the corner the map calls IQP-compatible, yet a permutation null that keeps each image's intensity histogram and destroys all spatial structure very nearly matches them, and the turbulence data behaves the same way, so at this encoding the map largely tracks the intensity histogram. Only the 10-bit crop pmf we train on separates from its permutation null in both coordinates, so we read a favourable map position as necessary rather than sufficient.

3.2 Why the reference protocol needs learned directions

Run sequentially on our low-variance latent block, Algorithm 1 produces noise (Fig. 1(b), second row; the Hamming-ordered variant is in App. E). Representability is not the cause, since the full 1,024-phase circuit fits a *single* aligned crop to correlation 0.99997 under the flat MMD. The protocol is. With the core frozen on one anchor, the 50 fixed latent coordinates reconstruct every other training image at correlation 0.47, because fixed Walsh directions act as an XOR-shift interference pattern that cross-fades pixels instead of moving lens structure.

3.3 Learned latent directions generalise

Equation (1) repairs this. Freezing θ^0 and Φ and optimising only z on 200 *held-out* images, the best arm at $N{=}100$, flat MMD at $M{=}4$, reaches mean correlation 0.968 against a nearest-training-image reference of 0.970, while the paper kernel reaches only 0.74–0.79. The gain comes from learning the directions and not from the extra parameters, since in the controlled $N{=}20$ sweep of Fig. 2(a) a trained Φ beats a random one of the same size by a wide margin. That sweep also shows that useful M stays small and collapses by $M{=}8$, once the required $\|z\|$ exceeds the model's coherence radius.

3.4 Paper-style comparison and data scaling

In Table 2 the $M{=}4$ paper-kernel arm wins PDF–JS at both training-set sizes and the $M{=}4$ flat-MMD arm wins the flux-aligned Conv2D-MMD^2 at both, on $2\times$ fewer parameters than the RBM and 36–39$\times$ fewer than the DCGAN, though no single arm wins both metrics. Two of the four IQP arms fall behind the DCGAN on PDF–JS at $N{=}11$ and two fall behind the RBM at $N{=}100$, and on raw Conv2D-MMD^2 every IQP arm loses at both sizes, an ordering that reverses once flux is aligned. The

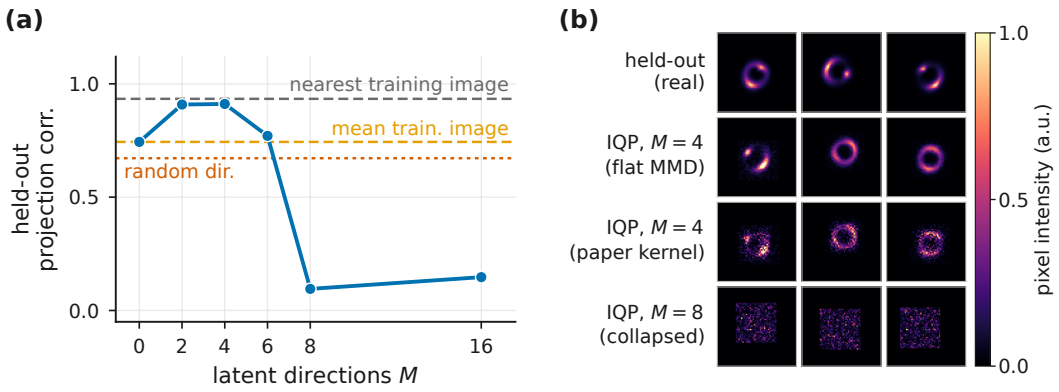


Figure 2: Panels (a) and (b) are different experiments. **(a)** Controlled latent-width sweep (N=20), held-out projection correlation against M, with its own reference levels (0.934 nearest training image, 0.744 mean training image, 0.672 random directions); the M=8 and 16 points are the best over a learning-rate sweep. **(b)** Round-3b arms at N=100, held-out crops on top, then M=4 under the flat MMD, M=4 under the paper kernel, and the collapsed M=8 arm, which was trained with KL and a z-penalty.

Table 2: Paper-style comparison on lens crops against the same two baselines. Lower is better and **bold** marks the best entry in a column. † runs Algorithm 1 on our low-variance latent block, learned-direction parameter counts are N=11 / N=100, MMD^2_{al} is the flux-aligned variant (post-hoc), and the last row is an empirical floor rather than a model.

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	MMD^2_{al}	PDF–JS	MMD^2	MMD^2_{al}
IQP, paper protocol (Alg. 1)†	1,024	0.00609	0.2031	0.1837	0.00697	0.2140	0.2233
IQP, fixed basis, joint	1,024	0.00133	0.1416	0.1141	0.00099	0.1041	0.0617
IQP, learned dir. M=4 (flat MMD)	5,164 / 5,520	0.00196	0.1134	**0.0279**	0.00239	0.0861	**0.0275**
IQP, learned dir. M=4 (paper kernel)	5,164 / 5,520	**0.00095**	0.1168	0.0679	**0.00092**	0.1189	0.0876
RBM [18]	11,010	0.00632	**0.0741**	0.1240	0.00195	0.0752	0.1144
DCGAN [13]	199,837	0.00183	0.2165	0.1241	0.00763	**0.0371**	0.0694
Held-out split-half (floor)	n/a	*0.00003*	*0.0040*	*0.0069*	*0.00003*	*0.0040*	*0.0069*

Table 3: Per-class generation at N=100 with the arm of Table 2, each run trained and scored inside one class, against the pooled arm. The N=11 runs are in App. F.

Training set	PDF–JS	MMD^2	MMD^2_{al}	proj. corr.
axion only	0.002050	0.09012	0.02346	0.9720
cdm only	0.002275	0.09783	0.01848	0.9686
no sub only	0.002001	0.1034	0.02773	0.9703
pooled	*0.002387*	*0.08607*	*0.02745*	*0.9684*

paper's data-scaling claim holds only in part, since our DCGAN is *better* at N=11 than at N=100 on PDF–JS, the opposite of the low-data GAN collapse the paper reports (Fig. 5, App. E).

Table 3 repeats the winning arm on each substructure class on its own. All three reach the pooled level, and all three still train at N=11 as well (App. F), while each class carries its own split-half floor, so a row compares with the pooled row rather than with the other classes.

4 Discussion and limitations

Nothing here is evidence of quantum advantage, because every circuit is simulated exactly at $n = 10$ qubits and the hardness results describe the model class at scale rather than our instances. All numbers are single seed, and the DCGAN's 3× swing in Fig. 5 sets the scale. Zero-learning and PCA controls match or exceed the IQP arms on some physical-quantity metrics.

References

[1] Stephon Alexander, Sergei Gleyzer, Evan McDonough, Michael W. Toomey, and Emanuele Usai. Deep learning the morphology of dark matter substructure. *The Astrophysical Journal*, 893(1):15, 2020.

[2] Stephon Alexander, Sergei Gleyzer, Hanna Parul, Pranath Reddy, Michael W. Toomey, Emanuele Usai, and Ryker Von Klar. Decoding dark matter substructure without supervision, 2020. arXiv:2008.12731. NeurIPS 2020 Workshop on Machine Learning and the Physical Sciences.

[3] Stephon Alexander, Sergei Gleyzer, Pranath Reddy, Marcos Tidball, and Michael W. Toomey. Domain adaptation for simulation-based dark matter searches with strong gravitational lensing. *The Astrophysical Journal*, 954(1):28, 2023. arXiv:2112.12121, doi:10.3847/1538-4357/acdfc7.

[4] Michael J. Bremner, Richard Jozsa, and Dan J. Shepherd. Classical simulation of commuting quantum computations implies collapse of the polynomial hierarchy. *Proceedings of the Royal Society A*, 467(2126):459–472, 2011.

[5] Michael J. Bremner, Ashley Montanaro, and Dan J. Shepherd. Average-case complexity versus approximate simulation of commuting quantum computations. *Physical Review Letters*, 117:080501, 2016.

[6] C. K. Chow and C. N. Liu. Approximating discrete probability distributions with dependence trees. *IEEE Transactions on Information Theory*, 14(3):462–467, 1968. doi:10.1109/TIT.1968.1054142.

[7] Julien Codsi and John van de Wetering. Classically simulating quantum supremacy IQP circuits through a random graph approach. *Physical Review A*, 111:012422, 2025.

[8] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012.

[9] Geoffrey E. Hinton. Training products of experts by minimizing contrastive divergence. *Neural Computation*, 14(8):1771–1800, 2002.

[10] Ali R. Khojasteh, Sylvain Laizet, Dominique Heitz, and Yin Yang. Lagrangian and Eulerian dataset of the wake downstream of a smooth cylinder at a Reynolds number equal to 3900. *Data in Brief*, 40:107725, 2022.

[11] Chen-Yu Liu, Leonardo Placidi, Eric Brunner, and Enrico Rinaldi. Toward generative quantum utility via correlation–complexity map, 2026. arXiv:2603.06440.

[12] ML4SCI Collaboration. DeepLense: Deep learning for strong gravitational lensing. `https://github.com/ML4SCI/DeepLense`, 2025. Model II dataset.

[13] Alec Radford, Luke Metz, and Soumith Chintala. Unsupervised representation learning with deep convolutional generative adversarial networks, 2015. arXiv:1511.06434.

[14] Joel Rajakumar, James D. Watson, and Yi-Kai Liu. Polynomial-time classical simulation of noisy IQP circuits with constant depth, 2024. arXiv:2403.14607. Also published in Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), pp. 1037–1056, doi:10.1137/1.9781611978322.30.

[15] Erik Recio-Armengol, Shahnawaz Ahmed, and Joseph Bowles. Train on classical, deploy on quantum: scaling generative quantum machine learning to a thousand qubits, 2025. arXiv:2503.02934.

[16] Erik Recio-Armengol and Joseph Bowles. IQPopt: Fast optimization of instantaneous quantum polynomial circuits in JAX, 2025. arXiv:2501.04776.

[17] Elad Schneidman, Susanne Still, William Bialek, and Michael J. Berry. Network information and connected correlations. *Physical Review Letters*, 91(23):238701, 2003.

[18] Tijmen Tieleman. Training restricted Boltzmann machines using approximations to the likelihood gradient. In *Proceedings of the 25th International Conference on Machine Learning (ICML)*, pages 1064–1071, 2008.

[19] Satosi Watanabe. Information theoretical analysis of multivariate correlation. *IBM Journal of Research and Development*, 4(1):66–82, 1960.

A Pipeline

Figure 3 summarises the encoding, the latent parameterisation and the evaluation.

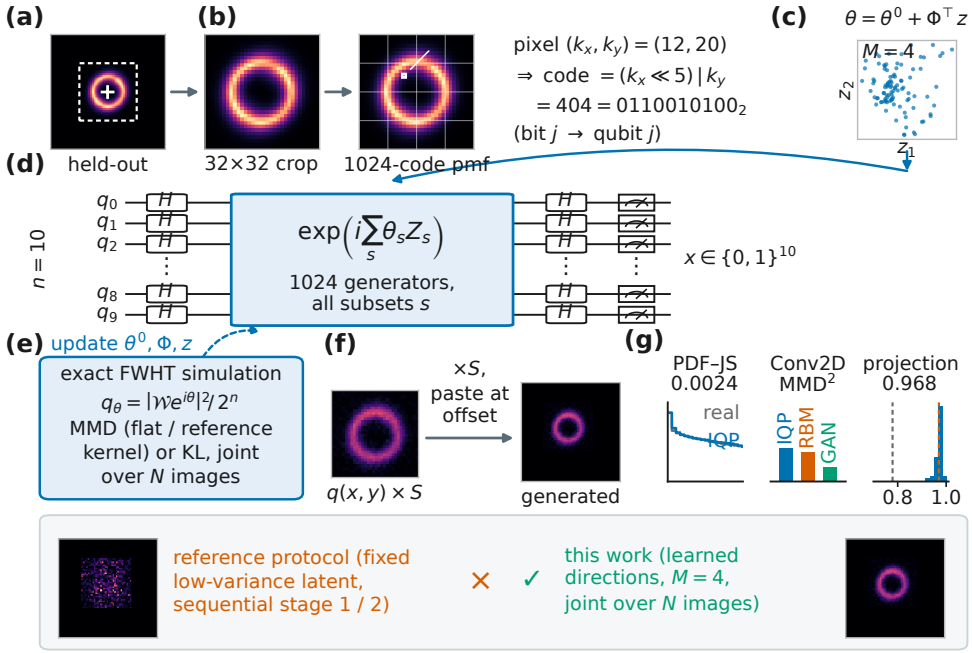


Figure 3: Pipeline. **(a)** A held-out 64×64 image, white cross at its intensity centroid and dashed box the 32×32 crop window. **(b)** The aligned crop read as a pmf over $1{,}024$ cells, where the example pixel $(k_x, k_y) = (12, 20)$ carries code $404 = 0110010100_2$ and bit j addresses qubit j. **(c)** The first two learned latent coordinates of the 100 training images at $M = 4$. **(d)** The IQP circuit, an H layer, one diagonal phase $\exp(i \sum_s \theta_s Z_s)$ over all $1{,}024$ subsets s, a second H layer and measurement. **(e)** Exact simulation through $q_\theta = |\mathcal{W} e^{i\theta}|^2 / 2^n$, trained jointly over the N images by MMD (flat or reference kernel) or by KL, the dashed arrow updating θ^0, Φ and z. **(f)** A trained $q(x, y)$ scaled by S and pasted at a sampled offset. **(g)** The intensity PDF of real against IQP samples (PDF–JS 0.0024), Conv2D-MMD^2 at $N{=}100$ (IQP 0.086, RBM 0.075, DCGAN 0.037), and projection correlations over 200 held-out images (mean 0.968). The bottom row contrasts a reference-protocol sample with an $M{=}4$ sample from this work.

B Replication of the reference paper's turbulence experiment

We attempted a direct replication on the reference paper's own dataset (2D cylinder-wake turbulence D2 [10], $n = 18$ qubits, 11 anchor snapshots). Under our pre-registered evaluation convention the result is **not reproduced**, since in the 11-snapshot regime the RBM and a 1.2M-parameter DCGAN beat our IQP on *both* of the paper's metrics. Two findings qualify this. Under a last-write-wins aggregation convention, never specified in the paper, our IQP's PDF–JS values do land close to the published ones (0.0322 and 0.0476 against 0.0238 and 0.0221), but the RBM still wins. Conv2D-MMD^2 is also saturated there, since ground truth shuffled in pixel space already scores 0.704, below every model in the paper's Table 2 (0.90–1.47), and the unseen-time synthesis task is solvable by linear interpolation (correlation 0.9998). Our anchor-fit ceiling (correlation 0.69 sampled, 0.77 exact) traces to a frozen θ_{lat} in stage 1 (≈ -0.08), a 5×10^4-sample pmf (≈ -0.04) and a 0.88–0.89 model-family ceiling, not to our implementation.

C Support mismatch (reference paper Fig. 3)

The reference paper's Fig. 3 claims (a) that the upper envelope of achievable MMD falls as the target's I_{QCLI} rises, because the support-induced floor shrinks, and (b) that the envelope drops overall as

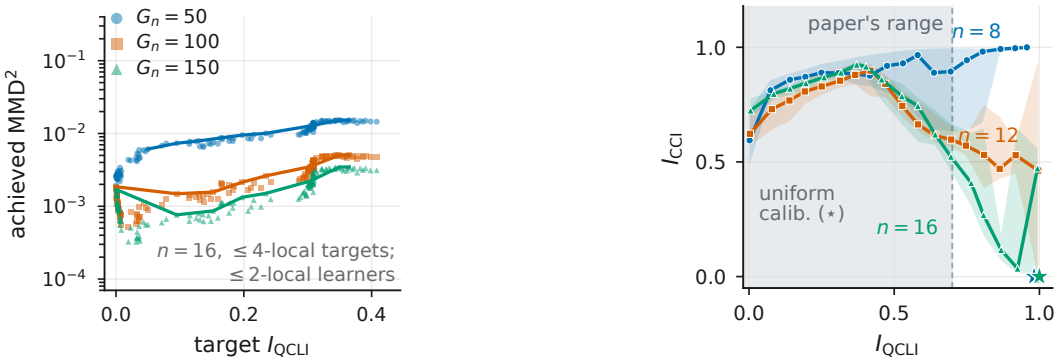


Figure 4: **(a)** Support-mismatch study at $n = 16$, with 140 target circuits ($\leq$ 4-local) spanning exact $I_{\mathrm{QCLI}} \in [0, 0.408]$ fitted by $\leq$ 2-local learners of 50, 100 and 150 gates (exact MMD, Adam, 5,000 steps). More gates lower the whole envelope (claim (b), reproduced), while within the reachable range the achieved MMD^2 *rises* with target I_{QCLI}, the opposite of claim (a). **(b)** Circuits optimised to maximise I_{QCLI} only, $n = 8, 12, 16$, 3 seeds and 2 gain settings per n, 3,333 steps each and about 20,000 logged points per n. I_{CCI} never enters the objective.

the learner gets more gates (Fig. 4(a)). Claim (b) reproduces, as the median achieved MMD^2 falls $9.57 \times 10^{-3} \to 2.15 \times 10^{-3} \to 1.45 \times 10^{-3}$ across the three learners. Claim (a) runs the other way inside the range we can reach, since the achieved MMD^2 *rises* as the target's exact I_{QCLI} goes from 0 to 0.41 (Spearman $\rho = +0.98/+0.90/+0.81$), a trend that also holds for a control trained on exact pmfs. The paper's premise still holds mechanistically ($\rho \leq -0.98$ for the Proposition II.2 floor proxy), but here the achieved MMD is dominated by trainability rather than by support mismatch. Our target family saturates at exact $I_{\mathrm{QCLI}} \approx 0.41$, and at 0–0.064 when estimated from 10^4 samples as the paper does, so the regime where claim (a) is asserted lies outside what we could construct.

D The I_{QCLI}–I_{CCI} ceiling

We also reproduced the experiment in which maximising I_{QCLI} alone raises I_{CCI} as a by-product (Fig. 4(b)). The coupling reproduces at low I_{QCLI} (Spearman $\rho = +0.83/+0.76/+0.95$ for $n = 8/12/16$ restricted to $I_{\mathrm{QCLI}} \leq 0.3$) but reverses above $I_{\mathrm{QCLI}} \approx 0.38$ for $n = 12, 16$ (global $\rho = -0.53, -0.57$). The reversal is forced by the definitions, because the exactly uniform distribution maximises I_{QCLI} (0.982/0.998/1.000 for $n = 8/12/16$) and has $I_{\mathrm{CCI}} = 0$. The reference paper's figure stops below $I_{\mathrm{QCLI}} \approx 0.7$, so a high I_{QCLI} need not indicate exploitable structure and can simply indicate proximity to uniform.

E Additional arms

Data scaling. At $N \in \{10, 30, 50, 70, 90, 100\}$ the learned-direction IQP arm is flat for $N \geq 30$ (raw Conv2D-MMD^2 0.082–0.087) and so is the RBM (0.066–0.089), while the DCGAN leads already at N=10 (0.044 against 0.120) and swings by 3× non-monotonically (0.113 → 0.037), as Fig. 5 shows.

Exact KL. KL needs the full simulated distribution and cannot be estimated from hardware shots, so we keep it out of Table 2 and record it here. Its M=2 arm, on 3,094 / 3,272 parameters, reaches PDF–JS 0.00175 at N=11 and 0.00302 at N=100, with raw Conv2D-MMD^2 0.1673 and 0.1356 and flux-aligned 0.0559 and

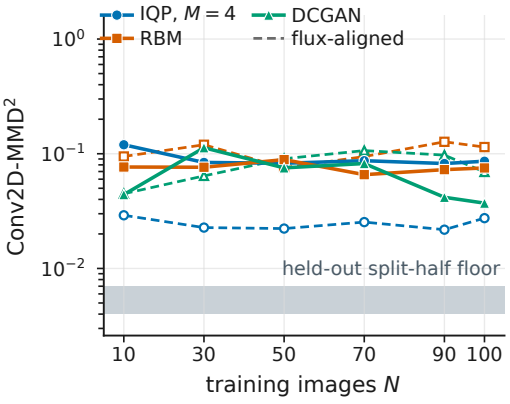


Figure 5: Conv2D-MMD^2 against training-set size, raw as solid lines and flux-aligned as dashed lines with open markers, over the band of the held-out split-half floor.

0.0866. The collapsed $M{=}8$ arm of Fig. 2(b) is also a KL arm.

Prior draws at $M{=}2$. All draws from the $M{=}2$ prior are rings of rms radius 9.40 ± 0.21 px against 9.21 ± 0.15 for real data, though under-dispersed in ellipticity (0.048 against 0.099).

Other arms. We ran 25 conditions, 16 IQP arms, the RBM and DCGAN baselines and seven zero-learning or PCA controls. The arms outside Table 2 (Hamming-ordered latent block, other objectives and priors, a pseudo-time conditional arm) and the controls are scored under the same metrics in the released tables. Trained jointly rather than sequentially, the Hamming-ordered latent block of the reference protocol reaches PDF–JS 0.00137 at $N{=}11$ and 0.00139 at $N{=}100$, which is consistent with the sequential protocol, rather than the choice of block, being what fails in Sec. 3.2. We did not run the Hamming block under the sequential protocol. Every IQP arm keeps a nearest-neighbour distance to the held-out set (3.6–5.9) far above the split-half floor (0.95), so none is memorising the training set, and the $M{=}8$ arm collapses (PDF–JS 0.0103, MMD^2 0.244 at $N{=}100$).

F Per-class samples

Figure 6 shows what the per-class runs of Table 3 generate, one row per class, from the arm of Table 2 retrained and scored inside each class with nothing else changed, and at $N{=}11$ the same three runs reach held-out projection correlations of 0.9232, 0.9238 and 0.8998 against the pooled arm's 0.9092. The rows look alike because the three classes differ in only a few per cent of the pixel variance, so we claim that each class can be generated on its own and not that the model separates them.

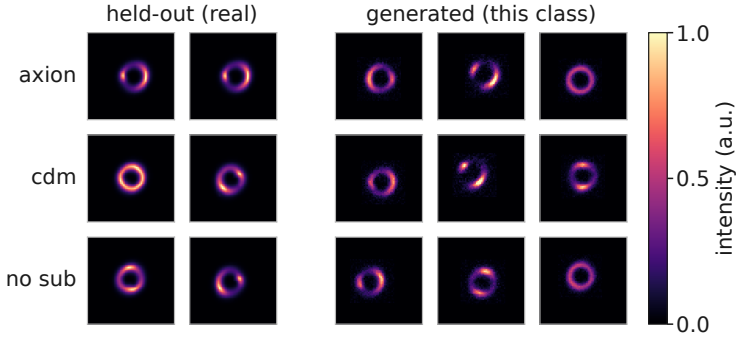


Figure 6: Per-class samples. Each row is one class, two held-out images followed by three generated by the arm trained on 100 images of that class alone, taken as the first two and the first three in index order with no selection, on the shared 0–1 colour scale.